

Lesson 19 More on Relations

1. Let $A = \{1,2,3\}$. For each relation R on A , determine if R reflexive, symmetric, antisymmetric, or transitive. If R is an equivalence relation, give the equivalence classes.
 - a) $R = \emptyset$
 - b) $R = \{(1,1)\}$
 - c) $R = \{(1,1), (1,2), (2,1)\}$
 - d) $R = \{(1,1), (1,2), (1,3)\}$
 - e) $R = \{(1,1), (1,2), (2,1), (2,2)\}$
 - f) $R = \{(1,1), (2,2), (2,3), (3,2), (3,3)\}$
 - g) $R = A \times A$
2. What are all possible equivalence relations on $A = \{1,2,3\}$?
3. Let $A = \{1,2,3,4\}$ and R be a relation on A given as $R = \{(1,1), (1,2), (1,3), (3,1)\}$.
 - a) Find the smallest relation S on A such that $R \subseteq S$ and S is reflexive.
 - b) Find the smallest relation S on A such that $R \subseteq S$ and S is symmetric.
 - c) Find the smallest relation S on A such that $R \subseteq S$ and S is transitive.
 - d) Find the smallest equivalence relation S on A such that $R \subseteq S$.
4. Prove that if $a = b \pmod{n}$ and $c = d \pmod{n}$, then $a + c = b + d \pmod{n}$ and $ac = bd \pmod{n}$.
5. Consider arithmetic on $\mathbb{Z}_8 = \{[0], [1], [2], \dots, [7]\}$. Find
 - a) $[5] + [6]$.
 - b) $[5] \times [6]$.
6. Let A be a set. Show that the subset relation on $\mathcal{P}(A)$ is not a total order.
7. Determine if each relation is a (i) partial order (ii) total order.
 - a) “ $<$ ” on $\mathbb{Z}$
 - b) The relation in Lesson 18 Exercises #5
 - c) “divides” on $\mathbb{N}$
 - d) **(Optional)** “implies” on the set of all statements

(Optional) To do division in $\mathbb{Z}_n$, we need to know what element of $\mathbb{Z}_n$ has a reciprocal. Let $n \in \mathbb{N}$ and $a \in \mathbb{Z}$. Prove that there is an integer b such that $ab \equiv 1 \pmod{n}$ if and only if $\gcd(a, n) = 1$.

Hint: Use Theorem 2 in Lesson 10.